



Arjun Nagar

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Bachelor of Technology in Electrical Engineering

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Portfolio Live

SUMMARY

A passionate Full-Stack Developer and Engineering student at NIT Kurukshetra with hands-on experience in building scalable web applications using the MERN stack. Proficient in C++, JavaScript, React.js, Node.js, Express.js, MongoDB, and modern web technologies, with a strong foundation in data structures and algorithms. Developed projects involving real-time communication, cloud integration, authentication, and AI-powered features. Eager to contribute technical expertise, learn emerging technologies, and build impactful software solutions in a collaborative engineering environment.

EDUCATION

Degree	Institute	Board / University	CGPA/Percentage	Year
B.Tech EE	National Institute of Technology, Kurukshetra	NIT Kurukshetra	7.71 (Till 5th Sem)	2023-2027
Senior Secondary	Modern Vidya Niketan Senior Secondary School	CBSE (Class XII), Faridabad	87.4%	2022

PROJECTS

• KineticCare – AI-Powered Physiotherapy Platform

2026

React.js, MediaPipe BlazePose, Three.js, Node.js, MongoDB, Tailwind CSS

Live | Github

- Engineered an AI-powered browser-based physiotherapy platform that performs **real-time human pose estimation** using **MediaPipe BlazePose**, coupled with **Three.js** to render an interactive 3D skeletal model for intuitive movement visualization.
- Developed a motion analysis engine using **3D vector mathematics** and joint-angle computations to accurately measure range of motion, achieving approximately $\pm 2.3^\circ$ **mean absolute error** during rehabilitation exercise tracking.
- Designed and implemented a **finite-state (hysteresis) machine** to automate repetition counting, posture validation, and exercise phase detection, significantly reducing false-positive detections caused by noisy pose estimation.
- Integrated the browser-native **Web Speech API** and **Web Audio API** to deliver low-latency voice guidance and real-time corrective feedback, creating an interactive hands-free rehabilitation experience.

• Schedura – AI-Powered Social Media Automation Platform

2025

React.js, Node.js, Express.js, MongoDB, Gemini API, Leonardo AI, JWT, Cloudinary

Live | Github

- Engineered a full-stack social media automation platform using the MERN stack, implementing secure **JWT authentication**, **role-based access control (RBAC)**, and modular **RESTful APIs** for scalable user and content management.
- Integrated **Gemini API** and **Leonardo AI** to automate AI-powered caption generation and image creation, significantly streamlining content creation workflows for social media publishing.
- Designed and implemented a scalable post scheduling pipeline supporting multi-platform publishing, asynchronous media uploads, and **Cloudinary**-based asset storage and delivery for efficient media management.
- Developed a real-time analytics dashboard to visualize engagement trends, publishing history, and campaign performance, enabling users to monitor and optimize their social media strategy through actionable insights.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, JavaScript
- **Frontend & Design:** HTML5, CSS3, React.js, Next.js, Tailwind CSS, Redux Toolkit, React Router, Responsive Web Design, Three.js
- **Backend & Databases:** Node.js, Express.js, RESTful APIs, JWT Authentication, OAuth 2.0, Bcrypt.js, Multer, MongoDB, PostgreSQL
- **Cloud, DevOps & Tools:** AWS S3, AWS SDK, Docker, Kubernetes, Git, GitHub, CI/CD, Vercel, Postman
- **Core Subjects:** DSA, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering, System Design Fundamentals

POSITIONS OF RESPONSIBILITY

- **Active Member**, National Service Scheme (NSS), NIT Kurukshetra

2023 – Present

- **Valorant Esports Player**, NIT Kurukshetra Esports Team

2024 – Present

ACHIEVEMENTS

- **14th Place**, VISIONARY Hackathon 2.0 (Mettl Xathon) – Led a multidisciplinary team to develop KineticCare with real-time 3D pose estimation
- **Coding Profile**, Maintained active competitive programming profile on Codolio with 500+ DSA problems solved *Ongoing*